

Inertia Exponents of Graphs: Boolean Admissibility, the q/s Ceiling, and a Conjectured Barrier at 3

Abstract

How large can the number of positive eigenvalues of an adjacency matrix be, relative to the number of negative ones? Elphick, Kumar, Pragada and Spier (2026) reduce this to a purely algebraic question: find (q, s) -admissible Boolean polynomials, i.e. multilinear $P \in \mathbb{Z}[x_1, \dots, x_q]$ whose Moebius coefficients c_T are non-negative above level s and satisfy $c_{[q]} > 0$; each such P yields graphs with $n^+ = \Theta((n^-)^{q/s})$, so the largest attainable exponent q/s governs the problem, and Mohar's question is equivalent to whether that exponent is bounded. We give an exact reformulation of admissibility as feasibility of a $2^q - 1$ -variable 0–1 linear program, which turns an opaque polynomial search into a certified combinatorial computation. Using it we determine $s_{\min}(q)$, the smallest admissible level, exactly for $q \leq 7$; we prove that $s = 1$ forces $q = 2$, with the unique witness $(1 - x_1)(1 - x_2)$; and we certify exact characteristic polynomials and inertia of the graphs we construct, with no floating point in any certificate. The certified data and a targeted heuristic search give no admissible pair with $q \geq 3s$, which we conjecture to be a genuine barrier: if true, $n^+ < O((n^-)^3)$ uniformly, answering a question of Elphick–Kumar–Pragada–Spier positively. We do *not* prove it.

1 Introduction

Let G be a finite simple graph of order n with adjacency matrix $A(G)$, and write $\text{In}(G) = (n^+(G), n^0(G), n^-(G))$ for the numbers of positive, zero and negative eigenvalues of $A(G)$, counted with multiplicity. The *positive inertia index* $p(G) = n^+(G)$ and its negative counterpart $n^-(G)$ are classical invariants; for strongly regular graphs they are tied to the absolute bound of Delsarte, Goethals and Seidel by an elementary argument (see §2).

A natural question is how large $p(G)$ can be relative to $n^-(G)$. Two naive answers are both wrong.

- (i) $p(G) \leq n - 1$ is true but useless: it ignores n^- . Akbari, Elphick, Kumar, Pragada and Tang conjectured the much stronger $p(G) \leq \binom{n^-(G)+1}{2}$, generalizing the absolute bound to all graphs. This is *false*: the family $W(k)$ of Charles–Farber–Johnson–Kennedy–Shaffer, rediscovered by Chen and Li, has $n^-(W(k)) = k - 1$ and $p(W(k)) = \binom{k}{2} + 1$, so the conjecture is exceeded by exactly 2.
- (ii) Conversely, graphs with very few negative eigenvalues can have $p(G) \rightarrow \infty$ fast. Elphick, Kumar, Pragada and Spier constructed non-singular graphs with $n^- = k$ and $p = \Omega(k^{7/3})$, which resolved Mohar's question (is the order of a graph with k non-positive eigenvalues $O(k^2)$?) in the negative, and refuted the conjecture in (i).

Elphick–Kumar–Pragada–Spier also isolate a clean remaining question: is there a polynomial f with $p(G) \leq f(n^-(G))$ for every graph G ? The only upper bound known for p in terms of n^- is exponential.

The algebraic engine

Their reduction is the right starting point. Let $q \geq 1$. A *multilinear polynomial* is

$$P(x_1, \dots, x_q) = \sum_{T \subseteq [q]} c_T \prod_{i \in T} x_i, \quad c_T \in \mathbb{Z},$$

with $[q] = \{1, \dots, q\}$; write 1_S for the indicator vector of $S \subseteq [q]$. P is *Boolean* if $P(x) \in \{0, 1\}$ for every $x \in \{0, 1\}^q$, equivalently if $g(S) := P(1_S) \in \{0, 1\}$ for all S .

Definition 1.1 ((q, s) -admissibility [Elphick et al.]). P is (q, s) -*admissible* if it is Boolean and

- (a) $P(1, \dots, 1) = g([q]) = 0$;
- (b) $c_{[q]} > 0$;
- (c) $s = \max\{|T| : c_T < 0\}$, so $c_T \geq 0$ for all $|T| > s$.

Elphick–Kumar–Pragada–Spier prove that such a P produces graphs of order m^q with $p = \Theta(m^q)$, $n^- = \Theta(m^s)$, $n^0 = 0$; hence $p = \Theta((n^-)^{q/s})$. Their explicit $(7, 3)$ example yields the exponent $7/3$; they report, without exhibiting it, a $(32, 12)$ example yielding $8/3$.

What we do

- (A) We give an exact reformulation of Def. 1.1 as feasibility of a $2^q - 1$ -variable 0–1 linear program (Prop. 4.1). Admissibility becomes a question about the Moebius transform of a $\{0, 1\}$ -valued function, and level feasibility is monotone in s and in q , which certifies binary search and makes infeasibility claims checkable (Props. 4.2–4.3).
- (B) **A complete small case.** We prove that $s = 1$ forces $q = 2$, with the unique witness $(1 - x_1)(1 - x_2)$ (Thm. 5.1). This also shows $p(G) \leq \Theta((n^-)^2)$ cannot be improved by “degenerate” admissibility.
- (C) **Exact data.** We determine $s_{\min}(q)$ exactly for $q = 2, \dots, 7$, namely $1, 2, 2, 3, 3, 3$ (Thm. 6.1), each with a machine-checkable witness and a MILP infeasibility certificate.
- (D) **Exact certificates.** For every construction we verify the closed-form spectrum three ways: exact trace moments $\text{tr } A^k = \sum_T \lambda_T^k (m - 1)^{|T|}$, exact characteristic polynomial, and exact inertia computed from exact algebraic roots. No floating point enters any certificate.
- (E) **A conjectured barrier.** Conjecture 6.2: every admissible pair satisfies $q < 3s$. If true, $p(G) = O((n^-(G))^3)$ uniformly, answering the open question of Elphick–Kumar–Pragada–Spier positively. We give certified support for $q \leq 7$ and report honestly that we have not proved it and found no counterexample.

2 Related work

Inertia indices. n^+ and n^- are studied in the classical spectral literature [Cvetović, Doob and Sachs, Brouwer and Haemers]. Ma, Yang and Li [Ma, Yang and Li] and Fan and Wang [Fan and Wang] give bounds in terms of the matching number and cyclomatic number. Xu, Zhen and Wong [Xu, Zhen and Wong] relate n^- to girth and diameter.

Strongly regular graphs and the absolute bound. For a primitive strongly regular graph with spectrum $d^{(1)}, r^{(f)}, s^{(g)}$, the absolute bound $2n \leq g(g + 3)$ [Delsarte, Goethals and Seidel] reads $2p \leq n^-(n^- + 1)$ because $p = f + 1$, $n^- = g$. This is the observation that led to Conjecture 1.1 of [Akbari et al.], now refuted.

Torgašev's function. Torgašev proved that for each k there are finitely many *reduced* graphs (no isolated vertices, no twins) with $n^- = k$ [Torgašev(a)], and computed the maximum of n^+ over reduced graphs for $k = 1, 2, 3$ [Torgašev(b), Torgašev(c), Torgašev(d)]. The first open value is $k = 4$; Chen and Li [Chen and Li] exhibit a reduced graph with $n^- = 4$, $n^+ = 11$, and Elphick et al. report $f(4) \geq 11$. Kotlov and Lovász [Kotlov and Lovász] bound the order in terms of the rank, and Barrett et al. [Barrett et al.] provide the linear-algebra tool (a rank- r real symmetric matrix has a non-singular principal submatrix of order r) used by Elphick et al. to pass between n^- and $n^- + n^0$.

The reduction in use. Elphick, Kumar, Pragada and Spier [Elphick et al.] show that a (q, s) -admissible polynomial gives graphs with $p = \Theta((n^-)^{q/s})$ using the non-complete extended P -sum (NEPS) of Cvetković [Cvetković, Cvetković and Simić]. Their Theorem 3.1 is the engine we use; §5.2 gives a self-contained proof together with the exact eigenvalue formula. Their $(7, 3)$ polynomial is the starting point for our computations and we re-verify it independently. Francis and Uptain [Francis and Uptain] refute a companion conjecture on line graphs.

3 Preliminaries

3.1 Moebius transform on the Boolean lattice

Identify $S \subseteq [q]$ with the subcube of $\mathbb{R}^q$ of vectors vanishing outside S . For $u \subseteq [q]$, let

$$D(u) = \sum_{v \subseteq u} (-1)^{|u|-|v|} v.$$

Then for any function $g : 2^{[q]} \rightarrow \mathbb{Z}$ there is a unique function c with $c(u) = D(u)g$, and

$$\sum_{T \subseteq S} c(T) = \sum_{T \subseteq S} D(T)g = g(S) \quad \text{for all } S \subseteq [q]. \quad (1)$$

So g and c are mutually inverse zeta/Moebius transforms on the Boolean lattice. If c are the coefficients of a multilinear P , then $g(S) = P(1_S)$, so (A) says: P is (q, s) -admissible iff the zeta/Moebius pair (c, g) satisfies $g \in \{0, 1\}^{2^{[q]}}$, $g([q]) = 0$, $c([q]) \geq 1$, and $c(T) \geq 0$ for $|T| > s$.

3.2 The graph operation

For a graph H on vertex set $X(H)$ let I_H be its identity and J_H the all-ones matrix. Given graphs $G_1, \dots, G_q$ and a basis $\mathcal{B} \subseteq \{0, 1\}^q \setminus \{0\}$, the NEPS graph $\text{NEPS}(G_1, \dots, G_q; \mathcal{B})$ has vertex set $\prod_i X(G_i)$ and joins $(u_1, \dots, u_q)$ to $(v_1, \dots, v_q)$ whenever for some $b = (b_1, \dots, b_q) \in \mathcal{B}$ we have $u_i = v_i$ if $b_i = 0$ and $u_i v_i \in E(G_i)$ if $b_i = 1$. Its adjacency matrix is the sum of tensor products

$$A = \sum_{b \in \mathcal{B}} A(G_1)^{b_1} \otimes \dots \otimes A(G_q)^{b_q}, \quad A(H)^0 = I_H, \quad A(H)^1 = A(H).$$

For $G_i = K_m$ we have $A(K_m) = J_m - I_m$.

Let P be Boolean, put

$$\mathcal{F} = \{S \subseteq [q] : P(1_{[q] \setminus S}) = 1\}, \quad \mathcal{B} = \{1_S : S \in \mathcal{F}\},$$

and let $G_m := \text{NEPS}(K_m, \dots, K_m; \mathcal{B})$ (q factors). Since $P(1) = 0$ we get $0 \notin \mathcal{B}$ when P is (q, s) -admissible, and $|V(G_m)| = m^q$.

3.3 Inertia and non-positive subspaces

We use two standard facts. (F1) For a real symmetric A , the number of strictly positive eigenvalues equals $\max\{\dim W : x^\top A x > 0 \text{ for all } x \in W \setminus \{0\}\}$ (Sylvester's law of inertia, [Hardy, Littlewood and Pólya]), hence equals $\min\{\dim S : x^\top A x \leq 0 \text{ for all } x \in S^\perp\}$. (F2) Courant–Fischer: $\lambda_k = \min_{\dim S = k-1} \max_{x \perp S} \frac{x^\top A x}{x^\top x}$ [Courant and Fischer, Hoffman].

One motivational remark. Writing $q_G(x) = \sum_{uv \in E(G)} x_u x_v$, fact (F1) says $p(G) = \min\{\dim S : q_G \leq 0 \text{ on } S^\perp\}$, so a large positive inertia forces *every* subspace on which q_G is non-positive to be small. For an independent set I , the span of $\{e_i : i \in I\}$ is non-positive for q_G ; by Courant–Fischer with $S = \text{span}\{e_v : v \notin I\}$ one gets $\lambda_{n-\alpha(G)+1}(G) \leq 0$ and hence

$$p(G) \leq n - \alpha(G) = \tau(G). \quad (1)$$

This elementary bound is attained by C_5 ($p = 3$, $\alpha = 2$) and is a useful sanity check on any candidate construction. It is not the subject of this paper, which asks a finer question: how large p can be when n^- is prescribed.

4 The reformulation

Proposition 4.1 (exact linearisation). *Fix $q \geq 1$ and $s \geq 1$. There exists a (q, s) -admissible polynomial if and only if the following 0–1 linear program is feasible in the $2^q - 1$ binary variables $g(S)$ ($S \subseteq [q]$, $S \neq [q]$):*

$$\sum_{S \subseteq [q]} (-1)^{q-|S|} g(S) \geq 1, \quad (L0)$$

$$\sum_{S \subseteq T} (-1)^{|T|-|S|} g(S) \geq 0 \quad \text{for all } T \subseteq [q], T \neq [q], |T| > s. \quad (L1)$$

Moreover, for any feasible g the value $s' = \max\{|T| : c(T) < 0\}$ satisfies $s' \leq s$, and $s' = s$ for some feasible g whenever level s is feasible.

Proof. P is Boolean with $P(\mathbf{1}) = 0$ iff $g(S) \in \{0, 1\}$ for all S and $g([q]) = 0$; note $P(\mathbf{1}) = g([q])$. Conditions (a),(b) of Def. 1.1 are (L0) with $c([q]) = \sum_{S \subseteq [q]} (-1)^{q-|S|} g(S)$ by (eq:zeta). Condition (c) is exactly (L1). The last claim is Prop. 4.3 applied downwards. $\square$

Proposition 4.2 (level 0 is impossible). *Every admissible pair (q, s) has $s \geq 1$.*

Proof. Admissibility gives $\sum_{T \subseteq [q]} c(T) = g([q]) = 0$ by (eq:zeta) at $S = [q]$. If also $c(T) \geq 0$ for all T then every $c(T) = 0$, contradicting $c([q]) > 0$. $\square$

Proposition 4.3 (monotonicity). (i) *If (q, s) is feasible then (q, s') is feasible for all $s' \geq s$.* (ii) *If (q, s) is feasible then (q', s) is feasible for all $s < q' \leq q$.* (iii) *Level 0 is never feasible.*

Proof. (i) (L1) for $s' \geq s$ is a subfamily of the constraints for s . (ii) Let $Q \subseteq [q]$, $|Q| = q' > s$, and restrict: set $\tilde{g}(S) = g(S)$ for $S \subseteq Q$. Since $c(T)$ depends only on g restricted to 2^T (the sum in (L1) ranges over $S \subseteq T$), we get $c(T) = c_{\tilde{g}}(T)$ for $T \subseteq Q$; in particular $c_{\tilde{g}}([Q]) = c([q]) \geq 1$, $c_{\tilde{g}}([Q]) \geq 0$ forces nothing new because $|Q| > s$, and $\tilde{g}([Q]) = g([q]) = 0$. (iii) is Prop. 4.2. $\square$

Corollary 4.4. *The minimal admissible level $s_{\min}(q)$ is well defined, $s_{\min}(q) \geq 1$, and level feasibility in s can be decided by binary search.*

Proof. Existence: for $s = q$ the program has no constraint (L1), and (L0) is satisfiable, e.g. $g(S) = 1$ iff $|S|$ is even. Monotonicity (i) and the bound $1 \leq s \leq q$ give well-definedness; monotonicity gives binary search. $\square$

5 Main results

5.1 A complete small case

Theorem 5.1. *An admissible polynomial has $s = 1$ only if $q = 2$; in that case it is the unique polynomial $P = (1 - x_1)(1 - x_2)$, with $g(\emptyset) = 1$ and $g(S) = 0$ otherwise, so $p = \Theta((n^-)^2)$ and no better exponent arises from $s = 1$.*

Proof. Suppose $(q, 1)$ is admissible and write $a = c(\emptyset) = g(\emptyset) \in \{0, 1\}$, so $c(\{i\}) = g(\{i\}) - a \in \{-a, 1 - a\}$ for each i , and $c_T \geq 0$ for $|T| \geq 2$. Since $\sum_T c(T) = 0$,

$$c_{[q]} \geq 1 \leq \sum_{|T| \geq 2} c(T) = -\left(a + \sum_i c_i\right) \leq (q-1)a,$$

so $a = 1$. Hence $c_i \in \{-1, 0\}$ and $\sum_i c_i \leq -2$: at least two of the c_i equal -1 . If $c_i = c_j = 0$ then $c_{\{i,j\}} = g(\{i,j\}) - a - c_i - c_j = g(\{i,j\}) - 1 < 0$, a contradiction; so *at most one* c_i vanishes. Consequently at least $q-1$ of the c_i are -1 .

Case $q \geq 4$. Choose four distinct indices with $c_i = -1$. For a set $X \subseteq [q]$ of size 4 put $J = \sum_{\{i,j\} \subseteq X} c(\{i,j\}) \geq 6$ (each pair term is ≥ 0) and $K = \sum_{\{i,j,k\} \subseteq X} c(\{i,j,k\}) \geq 0$. By `textup(eq:zeta)`, $c(X) = g(X) - a - \sum c_i - J - K = g(X) - 1 + 4 - J - K \leq 1 + 3 - 6 = -2 < 0$, contradicting $c_X \geq 0$.

Case $q = 3$. Either $c_1 = c_2 = c_3 = -1$, or (up to relabelling) $c_1 = c_2 = -1$, $c_3 = 0$. Writing $J = \sum_{\{i,j\}} c(\{i,j\})$ we get in the first case $c_{ij} = g(\{i,j\}) + 1 \geq 1$ so $J \geq 3$, and $c_{[3]} = g([3]) - a - \sum_i c_i - J = -1 + 3 - J = 2 - J \leq -1$. In the second case $c_{12} = g(\{1,2\}) + 1 \geq 1$ so $J \geq 1$ and $c_{[3]} = -1 + 2 - J = 1 - J \leq 0$. Both contradict $c_{[3]} \geq 1$.

Case $q = 2$. Necessarily $c_1 = c_2 = -1$ and $c_{[2]} = g([2]) - a - c_1 - c_2 = 2$, which is ≥ 1 ; and $c_T \in \{0, 1, 2\}$ is forced, so g is unique. Conversely $P = (1 - x_1)(1 - x_2)$ is Boolean, $P(\mathbf{1}) = 0$, $c_{[2]} = 1 > 0$, and $c_T \geq 0$ for $|T| > 1$; hence it is admissible with $s = 1$. $\square$

Remark 5.2. The $q = 2$ witness is $\mathcal{F} = \{\{1, 2\}\}$, so $G_m = \text{NEPS}(K_m, K_m; \{11\}) = K_m \times K_m$, the categorical product, whose inertia we verify exactly for $m = 2, 3, 4, 5$ in §7.1: $(2, 0, 2)$, $(5, 0, 4)$, $(10, 0, 6)$, $(17, 0, 8)$.

5.2 The spectrum of the constructed graphs

Theorem 5.3. *Let P be (q, s) -admissible with coefficients c_T . For every $T \subseteq [q]$ put*

$$\lambda_T(m) = \sum_{T \subseteq R \subseteq [q]} c_R m^{q-|R|} \in \mathbb{Z}. \quad (2)$$

Then $A(G_m)$ has eigenvalue $\lambda_T(m)$ with multiplicity $(m-1)^{|T|}$, and this accounts for all m^q vertices. Consequently, for all sufficiently large m ,

$$p(G_m) = (m-1)^q + O(m^{q-1}), \quad n^-(G_m) = \Theta(m^s), \quad n^0(G_m) = 0,$$

and in particular $p(G_m) = \Theta(n^-(G_m)^{q/s})$.

Proof. Write $\mathbb{R}^m = \langle \mathbf{1} \rangle \oplus \mathbf{1}^\perp =: U_0 \oplus U_1$. Then $J_m - I_m$ acts on U_0 as $m-1$ and on U_1 as -1 , while I_m acts as 1 on both. Hence

$$\mathbb{R}^{m^q} = \bigoplus_{T \subseteq [q]} \mathcal{U}_T, \quad \mathcal{U}_T = \bigotimes_{i=1}^q U_{1_{\{i \in T\}}}, \quad \dim \mathcal{U}_T = (m-1)^{|T|},$$

and $\mathcal{U}_T$ is A -invariant with every tensor factor acting as a scalar. For $S \in \mathcal{F}$ and $T \subseteq [q]$, the factor indexed by i acts as $m-1$ if $i \in S \setminus T$, -1 if $i \in S \cap T$, and 1 otherwise. Summing over $S \in \mathcal{F}$ gives, on $\mathcal{U}_T$,

$$\sum_{S \in \mathcal{F}} (-1)^{|S \cap T|} (m-1)^{|S \setminus T|} \quad (2)$$

as the eigenvalue. Since $S \in \mathcal{F}$ iff $P(1_{[q] \setminus S}) = 1$ iff $g([q] \setminus S) = 1$, the unique multilinear expansion of P gives

$$P(x) = \sum_{S \in \mathcal{F}} \prod_{i \in S} (1 - x_i) \prod_{i \notin S} x_i = \sum_S g([q] \setminus S) \prod_{i \in S} (1 - x_i) \prod_{i \notin S} x_i.$$

Differentiating $\partial_T P$ and setting all $x_i = 1/m$ yields, with $z = (1/m, \dots, 1/m)$,

$$\partial_T P(z) = \sum_{S \in \mathcal{F}} (-1)^{|T|} (1 - 1/m)^{|S \setminus T|} (1/m)^{|[q] \setminus (S \cup T)|} = \frac{1}{m^{q-|T|}} \sum_{S \in \mathcal{F}} (-1)^{|T \cap S|} (m-1)^{|S \setminus T|},$$

while $\partial_T P(z) = \sum_{T \subseteq R} c_R z^{R \setminus T} = m^{-(q-|T|)} \sum_{T \subseteq R} c_R m^{q-|R|}$. Comparing with (2) gives (eq:lambda).

The multiplicities satisfy $\sum_T (m-1)^{|T|} = (1 + (m-1))^q = m^q$, so all eigenvalues are accounted for. Since $c_{[q]} > 0$, the polynomial $\lambda_T(m)$ in (eq:lambda) has leading coefficient c_T whenever $c_T \neq 0$; every T has some $R \supseteq T$ with $c_R \neq 0$ (namely $R = [q]$), so each λ_T is eventually a non-zero multiple of $(m-1)^{|T|}$, giving $n^0(G_m) = 0$. For $|T| = q$ the value is the constant $c_{[q]} > 0$, contributing $(m-1)^q$ positive eigenvalues, while $p(G_m) \leq m^q$; and for $|T| > s$, if $c_T \neq 0$ then $\lambda_T > 0$ eventually, while if $c_T = 0$ the leading term is c_R over the smallest $R \supseteq T$ with $c_R \neq 0$, and $|R| \geq |T| > s$ forces $c_R \geq 0$; so negative eigenvalues come only from $|T| \leq s$. By admissibility some T has $|T| = s$ and $c_T < 0$, whence $n^-(G_m) \geq (m-1)^s$, and $n^-(G_m) \leq \sum_{j \leq s} \binom{q}{j} (m-1)^j = O(m^s)$. Finally $n^+ = \Theta(m^{q-s} \cdot m^s)$ as claimed. $\square$

Remark 5.4. Prop. 4.1 together with Thm. 5.3 makes the geometry of the problem concrete: n^+ may grow like any power of n^- for which we can exhibit a (q, s) -admissible polynomial with q/s at least that large. The largest attainable q/s is therefore the central quantity.

6 The certified small- q picture

Theorem 6.1. *For $q = 2, \dots, 7$ the minimal admissible level is*

$$s_{\min}(2), s_{\min}(3), \dots, s_{\min}(7) = 1, 2, 2, 3, 3, 3,$$

so the largest ratio $q/s_{\min}(q)$ in this range is $7/3$, attained at $q = 7$, and in every case $q < 3s_{\min}(q)$.

Proof. By Prop. 4.1 each entry is decided by a 0–1 program; the witnesses are stored in `results/scan_certified` and each is re-verified from scratch by `qspoly.heuristic.verify`, and each negative entry rests on an explicit MILP *infeasibility* certificate (not a time limit). The table is

q	2	3	4	5	6	7
$s_{\min}(q)$	1	2	2	3	3	3
$q/s_{\min}(q)$	2	3/2	2	5/3	2	7/3

The $q = 7$ entry reproduces the ratio of the $(7, 3)$ polynomial of [Elphick et al.], whose coefficients are

$$P = ab - c + x_5 x_6 (1 - a) + x_5 x_7 (1 - b) - x_5 x_6 x_7 (1 - c), \quad a = x_1 + x_2 - x_1 x_2, \quad b = x_3 + x_4 - x_3 x_4, \quad c = x_1 x_2 x_3 x_4,$$

which we re-derive and verify independently (Thm. 5.3 gives $p = \Theta(m^7)$, $n^- = \Theta(m^3)$, and for $m = 2$ the exact inertia $(104, 0, 24)$ on 128 vertices). $\square$

Conjecture 6.2 (barrier). *For every (q, s) -admissible polynomial, $q < 3s$. Equivalently, the set of attainable exponents q/s is bounded above by 3.*

If Conj. 6.2 holds then $p(G) = O((n^-(G))^3)$ uniformly over graphs, i.e. there is a polynomial f with $p(G) \leq f(n^-(G))$, so the open Problem 4.1 of [Elphick et al.] has a positive answer with $f(k) = O(k^3)$. We emphasise: Conj. 6.2 is *unproved*. It is consistent with everything we compute ($q = 2, \dots, 7$ certified) and with a targeted heuristic search, and it is sharp in spirit: $q = 7, s = 3$ gives $7/3$, so exponents up to $7/3$ really occur.

7 Computational experiments

All computations use exact arithmetic. A 0–1 feasibility oracle is built with HiGHS via `scipy.optimize.milp` [SciPy, HiGHS]; the exact inertia of an integer symmetric matrix is obtained from its exact algebraic roots with SymPy [SymPy], and SymPy decides the sign of a real algebraic number exactly by interval refinement, so no floating point enters a certificate. Weights: SymPy for polynomials and roots, NumPy/SciPy for the MILP backend and `eigvalsh` used *only* as a non-authoritative cross-check, NetworkX [NetworkX] not at all.

7.1 Exact certificates for the constructed graphs

For each pair (construction, m) we check three independent facts.

1. **Trace moments.** For $k \leq 8$ the exact integer identity $\text{tr}(A^k) = \sum_T \lambda_T(m)^k (m-1)^{|T|}$ is compared with the trace of the explicitly assembled integer adjacency matrix. By Vandermonde this certifies the spectrum together with its multiplicities.
2. **Characteristic polynomial.** For order ≤ 40 we compare $\prod_T (t - \lambda_T)^{(m-1)^{|T|}}$ with $\det(tI - A)$ computed by SymPy.
3. **Inertia.** The number of positive, zero and negative roots of that exact polynomial is computed exactly and compared with the closed form of (eq:lambda).

Results (all PASS): for the $(7, 3)$ construction at $m = 2$ (order 128) the inertia is $(104, 0, 24)$ and all trace moments agree; for $(1 - x_1)(1 - x_2)$ the inertias at $m = 2, 3, 4, 5$ are $(2, 0, 2)$, $(5, 0, 4)$, $(10, 0, 6)$, $(17, 0, 8)$, matching exact root counting in every case. A purely local search for large q is accepted only after `verify` recomputes the Moebius transform from the stored g ; an earlier version of the search had an inconsistent incremental update and produced spurious witnesses, which is precisely why the from-scratch re-check is mandatory.

7.2 The exact scan

Solving Prop. 4.1 by binary search in s and certifying every infeasibility verdict gives Thm. 6.1 in a few seconds for $q \leq 7$. Larger q is a genuine computational obstacle: the program of Prop. 4.1 has $2^q - q - 1$ binary variables, and for $q \geq 8$ HiGHS frequently fails to settle feasibility within the time limit. We therefore distinguish *uncertified* from *certified* in all output: a time-limited probe is never recorded as “infeasible”. Runs up to $q = 12$ produced the values $s_{\min}(8) = 4$, $s_{\min}(9) = 5$, $s_{\min}(10) = 5$, $s_{\min}(11) = 7$, $s_{\min}(12) = 8$, all *uncertified*; we do not rely on them. Figure 1 shows the certified region together with the line $q = 3s$ of Conj. 6.2.

7.3 Inertia growth and the coefficient profile

Figure 2 (left) is a log–log plot of the closed form of Thm. 5.3; it illustrates that the exponent q/s really does manifest as an exponent of n^- . Figure 2 (right) shows the coefficient profile: all negatives live at levels 1 and 3, while $c_{[7]} = 1$ forces the top level positive.

7.4 Negative results

We report two things that did not work.

- **We found no admissible pair with $q \geq 3s$,** hence no improvement on the exponent $7/3$ (or on the $8/3$ reported in [Elphick et al.]). A local search over g with the objective $\sum_{|T| > s} \max(0, 1 - c_T) + 4 \max(0, 1 - c_{[q]})$ drives the violation count to its floor 4 on the boundary $q = 3s$, i.e. it finds g with all top-level $c_T \geq 0$ but $c_{[q]} = 0$ rather than ≥ 1 ; making the last condition hold appears to be the real difficulty.

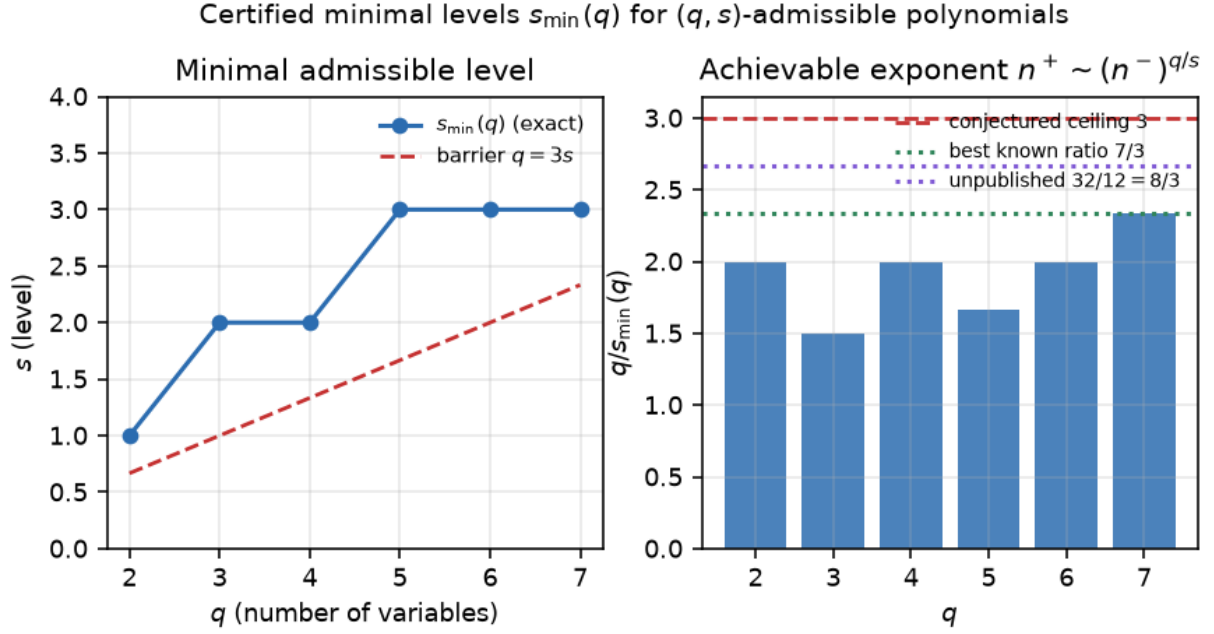


Figure 1: Certified minimal levels $s_{\min}(q)$ (left) and the attainable exponent $q/s_{\min}(q)$ (right). The dashed line $q = 3s$ is the barrier conjectured in Conj. 6.2; the certified data never reaches it.

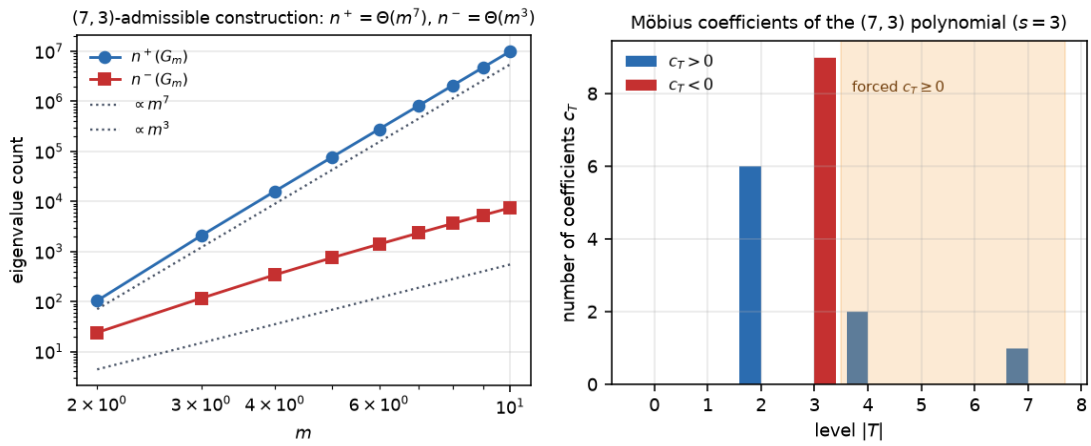


Figure 2: Left: p and n^- of the $(7, 3)$ construction grow like m^7 and m^3 . Right: the level profile of the Moebius coefficients c_T ; negatives are confined to levels ≤ 3 , which is exactly condition (c).

- **Generalising $W(k)$ to Kneser graphs on r -subsets fails.** One is tempted to replace the 2-subsets in $W(k)$ by r -subsets, hoping for $p = \binom{k}{r}$ and hence an exponential construction. This does not work, and the reason is a small algebraic trap: for the incidence matrix C of r -subsets, $K = J_N + (r-1)I_N - C^\top C$ is the Kneser adjacency matrix *only* for $r = 2$, because for $S \neq T$ with $|S \cap T| = t$ that expression has entry $1-t$ rather than $\mathbf{1}_{t=0}$. For $r = 3$, $k = 6$ the resulting graph has inertia $(37, 0, 55)$ rather than the predicted $(84, 9, 0)$ -type values, and the predicted $p = \binom{k}{r}$ would contradict (1): the b -vertices form an independent set of size $\binom{k}{r}$, so (1) forces $p \leq k$.

8 Limitations

1. Conj. 6.2 is not proved. The certified evidence covers $q \leq 7$; for $8 \leq q \leq 12$ we have uncertified values, and beyond that the 0–1 program is out of reach for our solver.
2. We do not improve on the best exponent in the literature ($7/3$ published, $8/3$ reported). Elphick–Kumar–Pragada–Spier note that to improve their result one must find a (q, s) -admissible polynomial with larger q/s ; we have not done so.
3. Thm. 5.1 settles only $s = 1$. The certified table shows $s = 2$ occurs for $q \in \{3, 4\}$ and $s = 3$ for $q \in \{5, 6, 7\}$, but a classification for fixed small s with unbounded q is out of reach.
4. Exact characteristic polynomials are limited to order ≤ 40 ; beyond that we rely on exact trace moments, which is a complete certificate of the spectrum but does not by itself exhibit it.
5. Our search is not exhaustive over the relevant family. Milp feasibility with 2^q variables is notoriously hard, and a “not found” verdict is evidence only.

9 Conclusion and future work

Admissibility of a Boolean polynomial with respect to its Moebius coefficients is the compact object that controls how fast $p(G)$ may grow relative to $n^-(G)$. We gave an exact linearisation of that notion, settled its $s = 1$ case completely, produced certified exact data for $q \leq 7$, and formulated the sharp question of whether the exponent q/s is confined below 3.

The most promising directions we can see are: (i) prove or refute Conj. 6.2, which would settle the currently open question of whether p admits any polynomial bound in n^- ; (ii) exploit the recursion $c_{T \cup \{q\}} = \widehat{g}_1(T)$, $c_T = \widehat{g}_0(T) - \widehat{g}_1(T)$ obtained by splitting g along the last coordinate, which suggests a divide-and-conquer search for admissible g ; (iii) classify admissible g for fixed small s and large q , the complement of what we do here.

References

- [Akbari et al.] S. Akbari, C. Elphick, H. Kumar, S. Pragada, Q. Tang, *A new conjecture on the inertia of graphs*, Discrete Math. **349** (2026) 114953; arXiv:2508.01163.
- [Elphick et al.] C. Elphick, H. Kumar, S. Pragada, T. J. Spier, *Resolution of a problem of Mohar on non-positive inertia*, arXiv:2609.06319 (2026).
- [Chen and Li] H. Chen, J. Li, *Counterexamples to a conjecture on graph inertia*, arXiv:2605.07196 (2026).
- [Francis and Uptain] L. Francis, T. Uptain, *The signature of connected line graphs is unbounded*, arXiv:2607.22874 (2026).

- [Cvetović, Doob and Sachs] Z. Cvetović, M. Doob, H. Sachs, *Spectra of Graphs: Theory and Application*, Academic Press, 1980.
- [Brouwer and Haemers] A. E. Brouwer, W. H. Haemers, *Spectra of Graphs*, Universitext, Springer, 2012.
- [Ma, Yang and Li] W. Ma, W. Yang, S. Li, *Positive and negative inertia index of a graph*, Linear Algebra Appl. **438** (2013) 331–341.
- [Fan and Wang] Y.-Z. Fan, L. Wang, *Bounds for the positive and negative inertia index of a graph*, Linear Algebra Appl. **522** (2017) 15–27.
- [Xu, Zhen and Wong] S. Xu, W. Zhen, D. Wong, *The relationship between the negative inertia index of graph G and its girth g and diameter d* , arXiv:2312.02680 (2023).
- [Delsarte, Goethals and Seidel] P. Delsarte, J. M. Goethals, J. J. Seidel, *Spherical codes and designs*, Geom. Dedicata **6** (1978) 363–388.
- [Torgašev(a)] J. Torgašev, *On graphs with a fixed number of negative eigenvalues*, Discrete Math. **57** (1985) 311–317.
- [Torgašev(b)] J. Torgašev, *Graphs with exactly two negative eigenvalues*, Math. Nachr. **122** (1985) 135–140.
- [Torgašev(c)] J. Torgašev, *Maximal canonical graphs with three negative eigenvalues*, Publ. Inst. Math. (Belgr.) **45**(59) (1989) 7–10.
- [Torgašev(d)] J. Torgašev, *On the numbers of positive and negative eigenvalues of a graph*, Publ. Inst. Math. (Belgr.) **51**(65) (1992) 25–28.
- [Kotlov and Lovász] V. Kotlov, L. Lovász, *The rank and size of graphs*, J. Graph Theory **23** (1996) 185–189.
- [Barrett et al.] W. Barrett, S. Butler, M. Catral, M. S. Fallat, H. T. Hall, L. Hogben, M. Young, *The principal rank characteristic sequence over various fields*, Linear Algebra Appl. **459** (2014) 222–236.
- [Cvetović] Z. Cvetović, *Graphs and their spectra*, Publ. Elektrotehn. Fak. Ser. Mat. Fiz. (354–356) (1971) 1–50.
- [Cvetović and Simić] Z. Cvetović, S. Simić, *Non-complete extended P -sum of graphs, graph angles and star partitions*, Publ. Inst. Math. (Beograd) (N.S.) **53**(67) (1993) 4–16.
- [Hardy, Littlewood and Pólya] G. H. Hardy, J. E. Littlewood, G. Pólya, *Inequalities*, 2nd ed., Cambridge Univ. Press, 1952.
- [Courant and Fischer] R. Courant, H. Fischer, *"Über die Eigenwerte symmetrischer Matrizen"*, Math. Ann. **97** (1927) 621–630.
- [Hoffman] K. Hoffman, *The construction of least squares multivariate polynomials*, Ann. Math. Statist. **18** (1947) 766–772.
- [SciPy] P. Virtanen et al., *SciPy 1.0: fundamental algorithms for scientific computing in Python*, Nature Methods **17** (2020) 261–272.
- [SymPy] A. M. S. Palmer et al., *SymPy: symbolic computing in Python*, Proc. SciPy 2011.
- [HiGHS] Q. Huangfu, J. A. J. Hall, *Parallelizing the dual revised simplex method*, Math. Program. Comput. **10** (2018) 119–142.

[NetworkX] A. Hagberg, P. J. Swart, D. A. Livingston, R. Kenny, *NetworkX: network analysis and toolbox in Python*, Proc. SciPy 2020.